Efficient Monte Carlo Pricing of Barrier Options under the OUSV Model

A Variance Reduction Approach

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Research Question

This thesis studies the Monte Carlo pricing of barrier options under the OUSV model, with a numerical focus on discretely monitored up-and-in and up-and-out call options.

Main question

After fixing the option type, barrier level and OUSV simulation scheme, which Monte Carlo method — Plain MC or a variance reduction technique — provides the best trade-off between statistical precision and computational cost?

Barrier Options

Barrier options are path-dependent derivatives whose payoff depends not only on the terminal value S_T, but also on whether the underlying reaches a predetermined barrier B before maturity.

Why use them? They allow firms and investors to obtain protection or market exposure only when a specific price level is reached. By giving up protection or exposure outside that scenario, they require a lower premium than for the corresponding vanilla option.

Up-and-in call discounted payoff

$$Y^{in} = e^{-rT}(S_T - K)^+ \mathbf{1}_{\{\tau_B \leq T\}}$$

Up-and-out call discounted payoff

$$Y^{out} = e^{-rT}(S_T - K)^+ \mathbf{1}_{\{\tau_B > T\}}$$

$$\tau_B = \min\{t_i : S_{t_i} \geq B\}$$

The OUSV Model

The asset price is driven by a latent Ornstein–Uhlenbeck volatility factor.

$$dS_t = rS_t dt + S_t v_t \left(\rho dW_t^{(2)} + \sqrt{1-\rho^2} dW_t^{(1)} \right)$$

$$dv_t = k(\theta - v_t)dt + \xi dW_t^{(2)}$$

- v_t is the latent volatility factor.
- $V_t = v_t^2$ is the variance process.
- The instantaneous volatility is $|v_t|$.
- ρ links asset-price shocks and volatility shocks.

Monte Carlo Estimation

The option price under a risk-neutral probability measure is:

$$V_0 = \mathbb{E}^Q[Y]$$

The plain Monte Carlo estimate is:

$$\widehat{V}_M = \frac{1}{M} \sum_{j=1}^{M} Y_j$$

The estimated standard error is:

$$\widehat{SE}(\widehat{V}_M) = \frac{s_Y}{\sqrt{M}}$$

The empirical efficiency indicator is:

$$Efficiency_m(B) = t_m(B)\, SE_m(B)^2$$

Lower values indicate a better time-variance trade-off.

Variance Reduction Techniques

The variance reduction techniques modify the random variable used in the Monte Carlo sample average while preserving the target option value.

Method	**Random variable averaged**	**Main idea**
Plain MC	Y	Average realizations of the discounted payoff Y.
Antithetic Variables	$Y^{AV} = \frac{Y + \widetilde{Y}}{2}$	Average the payoffs obtained from the original and antithetic paths.
Control Variates	$Y^{CV} = Y + \alpha(X - \mu_X)$	Use the corresponding vanilla call payoff as a control variable.
Conditional MC	$Y^{in,CMC} = e^{-r\tau_B} C(S_{\tau_B}, K, T - \tau_B)\mathbf{1}_{\{\tau_B \leq T\}}$	Replace the remaining random payoff after barrier activation with its conditional expected value, i.e. the residual vanilla call price. For up-and-out calls, use in-out parity.

For Control Variates:

$$X = e^{-rT}(S_T - K)^+, \qquad \mu_X = \mathbb{E}^Q[X] = C_0^{Fourier}.$$

OUSV Simulation Schemes

Three schemes are used to generate OUSV paths, meaning paths of both S_t and v_t.

Scheme	Main feature	Cost
Euler	Local discretization	Low
VHP	Exact OU transition for v_t	Low
Inverse Gaussian	Simulates integrated quantities	High

Key trade-off: more refined schemes, such as the Inverse Gaussian Scheme, can reduce discretization bias, but usually require more computational time.

Numerical Setting

The numerical analysis considers up-and-in and up-and-out calls for different barrier levels:

$$B \in \{105, 110, 120, 130, 140, 150, 160, 190\}$$

Parameter	Value
S_0	100
K	100
r	0.05
T	1
v_0	0.25
k	8

Parameter	Value
θ	0.25
ξ	0.30
ρ	-0.60
N	504
M	10000
M_{pilot}	1000

Results: Up-and-In Calls

For up-and-in calls, Control Variates is generally the most efficient variance reduction technique.

Barrier	Euler	VHP	IG
105	Control Var	Control Var	Control Var
110	Control Var	Control Var	Control Var
120	Control Var	Control Var	Control Var
130	Control Var	Control Var	Control Var
140	Control Var	Control Var	Control Var
150	Control Var	Control Var	Control Var
160	Control Var	Control Var	Control Var
190	Plain MC	Plain MC	Plain MC

Interpretation: for low and medium barriers, the up-and-in payoff remains strongly related to the vanilla payoff. This makes the control variate particularly effective. At the highest barrier, the knock-in event becomes rare and Plain MC becomes competitive.

Results: Up-and-Out Calls

For up-and-out calls, the most efficient Monte Carlo method depends more strongly on the barrier level.

Barrier	Euler	VHP	IG
105	Plain MC	Plain MC	AV
110	Control Var	Plain MC	AV
120	Control Var	Control Var	Plain MC
130	Control Var	Control Var	Plain MC
140	Control Var	Control Var	AV
150	Control Var	Control Var	AV
160	Control Var	Control Var	Control Var
190	Control Var	Control Var	Control Var

Interpretation: for low barriers, many paths are knocked out early, so Plain MC or Antithetic Variables can be efficient. For high barriers, the up-and-out payoff becomes closer to the vanilla payoff, so Control Variates becomes more effective.

Conclusions

- No variance reduction technique is uniformly the best.
- Control Variates performs very well for up-and-in calls and for high-barrier up-and-out calls.
- Plain MC and Antithetic Variables remain competitive for low-barrier up-and-out calls.
- Conditional MC reduces the standard error, but repeated Fourier evaluations make it computationally expensive.

Final message: In the numerical setting considered, the most efficient Monte Carlo method depends jointly on the option type, the barrier level and the OUSV simulation scheme.